



School of Engineering & Technology

Department of Civil Engineering

B.Tech. - Civil Engineering

Semester I & Semester II

AY 2023-24 onwards

Semester I

Theory/ Practical	Sl. No	Course Type	Course Code	Course Name	Hours per week			Credit	Mark	
					L	T	P		CA	FA
T	1.	BSC		Physics	3	1	0	4	40	60
P	2.	BSC		Physics Laboratory	0	0	2	1	40	60
T	3.	BSC		Mathematics - I	3	1	0	4	40	60
T	3.	BSC		Biology for Engineers	2	0	0	2	40	60
T	4.	ESC		Basic Electrical Engineering	3	1	0	4	40	60
P	5.	ESC		Basic Electrical Engineering Laboratory	0	0	2	1	40	60
T/P	6.	ESC		Engineering Graphics and Design	1	0	4	3	40	60
P	7.	ESC		Design Thinking & Idea Lab	0	0	2	1	100	00
P	8.	ESC		Manufacturing Practice Workshop-I	0	0	2	1	40	60
P	9.	AU		Sports/Yoga/NSS/NCC/Activities	0	0	2	0	100	00
Total					12	3	14	21		

Semester II

Theory/ Practical	Sl. No	Course Type	Course Code	Course Name	Hours per week			Credit	Mark	
					L	T	P		CA	FA
T	1.	BSC		Chemistry	3	0	0	3	40	60
P	2.	BSC		Chemistry Laboratory	0	0	2	1	40	60
T	3.	BSC		Mathematics - II	3	1	0	4	40	60
T	4.	ESC		Programming for Problem Solving	2	0	0	2	40	60
P	5.	ESC		Programming for Problem Solving Laboratory	0	0	4	2	40	60
T/P	6.	HSMC		English for Technical Communication	2	0	2	3	40	60
T	7.	AU		Universal Human Values	2	1	0	3	00	100
P	8.	ESC		Manufacturing Practice Workshop-II	0	0	2	1	40	60
T	9.	HSMC		Japanese	2	0	0	2	50	50
Total					14	2	10	21		

Internal Member

External Member/BoS



BSC	CHEMISTRY	L	T	P	C
		3	0	0	3
Prerequisite: Knowledge of Chemistry at (10+2) level					
Course Objectives:					
1. To provide knowledge of molecular orbital theory along with electronic configuration on the basis of Schrodinger wave equation for simple homonuclear and heteronuclear diatomic molecules. 2. To analyze different compounds with the help of different spectroscopic techniques. 3. To make students aware of the relationships between different thermodynamics properties with reference to chemical systems. 4. To provide brief idea about sustainable chemistry. 5. To provide an insight into different types of fuel and applications of various engineering materials.					
Course Outcome:					
After successful completion of the course, the students will be able					
CO1: To analyse microscopic chemistry in terms of atomic and molecular orbitals.					
CO 2: To apply the fundamental principles and applications of different spectroscopic techniques.					
CO 3: To explain bulk properties and processes using thermodynamic considerations.					
CO 4: To explain sustainable chemistry along with its applications.					
CO 5: To explain the chemistry of different types of fuel and engineering materials.					
Module 1: ATOMIC AND MOLECULAR STRUCTURE					6 hours
Wave property of matter, Schrodinger's wave equation, wave function, radial and angular wave functions, Eigen function, Eigen value, Particle in an one dimensional box and quantization of energy, Three dimensional potential box and degeneracy of energy states, Molecular Orbital Theory – Applications of MO Theory in diatomic molecules (N ₂ , O ₂ , NO and CO)					
Module 2: SPECTROSCOPIC TECHNIQUES AND APPLICATIONS					7 hours
Principle of spectroscopy, principle and applications of UV – Visible spectroscopy, infra-red spectroscopy, applications of nuclear magnetic resonance spectroscopy, atomic absorption spectroscopy and flame photometry. Fluorescence and its applications in medicine.					
Module 3: USE OF FREE ENERGY IN CHEMICAL EQUILIBRIA					6 hours
Entropy and randomness, Entropy change in reversible and irreversible processes, free energy, free energy as a criteria for spontaneity of a process, relationship between free energy change and entropy change, Dependence of Gibbs free energy on temperature and pressure, free energy and EMF, Cell potentials, the Nernst equation and applications.					
Module 4: GREEN AND SUSTAINABLE CHEMISTRY					5 hours
Principles of green chemistry, Idea of green synthesis, Carbon footprint and sequestration, Carbon trading. Brief idea of alternative solvents–Water, ionic liquids, supercritical fluid system (Carbon dioxide), Waste management: Solid, electronic & industrial wastes, Waste management procedures and relevant standards.					
Module 5: CORROSION AND ITS PREVENTION					5 hours
Definition, causes, effects, Dry or chemical corrosion and wet or electrochemical corrosion - their mechanisms. Types of electrochemical corrosion (Differential aeration, Galvanic, Concentration cell), Typical electrochemical corrosion like Pitting, Waterline. Factors affecting corrosion, passivity, Protection against corrosion.					
MODULE 6 : FUEL AND COMBUSTION					7 hours
Classification of fuel, calorific value, characteristics of a good fuel, determination of calorific value of fuel using the Bomb Calorimeter, calorific value from Dulong's formulae, classification of coal, proximate and ultimate analysis of coal, fractional distillation of petroleum, cracking,					



thermal and catalytic cracking, Refining of gasoline, Reforming, knocking, octane rating of fuel, Chemical structure of knocking, Antiknocking agents, Diesel fuel, cetane number, additives for diesel fuel,

MODULE 7: ADVANCED ENGINEERING MATERIALS**9 hours**

Cement - Cement and its classification, Portland cement, raw materials, manufacture, and its setting and hardening. Refractory materials - Definition, classification into acidic, basic and neutral refractories and their uses. Lubricants – Definition and function of lubricants, classification, additives for lubricants.

Total hours**45 hours****Text Book(s)**

1. Engineering Chemistry - Jain & Jain, Dhanpat Rai & Company.
2. A Text Book of Engineering Chemistry – Dr. Sunita Rattan, K. Kataria & Sons.
3. A Text Book of Engineering Chemistry - Dr. Rajashree Khare, S. K. Kataria & Sons.

Reference Books

1. Physical Chemistry, P. W. Atkins, Oxford.
2. Concise Inorganic Chemistry, J. D. Lee, Blackwell Science
3. Fundamentals of Molecular Spectroscopy, C. N. Banwell, E. M. Mc Cash, Tata McGraw – Hill.
4. Principles of Physical Chemistry, Puri, Sharma, Pathania, Shoban Lal Nagin Chand & Co.
5. Spectroscopy of Organic Compounds, P. S. Kalsi, Wiley Eastern.



BSC	CHEMISTRY LABORATORY	L	T	P	C
		0	0	2	1
Pre-requisite: Knowledge of Chemistry at (10+2) level					
Course Objectives:					
1. To make students familiar with different quantitative analysis. 2. To enable students carry out experiments using theoretical knowledge. 3. To provide knowledge of different properties of liquids by experimental methods.					
Course Outcome:					
After successful completion of the course, the students will be able					
CO1: To conduct quantitative analysis of a given substance by using different types of Volumetric titrations.					
CO2: To apply theoretical knowledge to carry out different experiments skillfully.					
CO3: To learn the physical properties like surface tension and viscosity of liquids by conducting the experiments.					
List of Experiments					
1. Estimation of hardness of water by a standard solution of EDTA 2. Estimation of Fe^{2+} by a standard solution of KMnO_4 3. Estimation of Cu^{2+} by a standard solution of $\text{Na}_2\text{S}_2\text{O}_3$ 4. Conductometric titration between strong acid and strong alkali 5. pH-metric titration between strong acid and strong alkali 6. Determination of surface tension of a liquid at room temperature w. r. t water by drop number method using stalagmometer. 7. Determination of coefficient of viscosity of a given solution at room temperature by Ostwald's Viscometer. 8. Preparation of potash alum, $[\text{K}_2\text{SO}_4.\text{Al}_2(\text{SO}_4)_3.2\text{H}_2\text{O}]$					
List of Equipments					
• Ostwald's viscometer • Stalagmometer • Conductivity meter • pH meter					
Total hours: 15 hours					
Text Book(s)					
1	Laboratory Manual on Engineering Chemistry by S. K. Bhasin and Sudha Rani.				
2	Practical Engineering chemistry by Sunitha and Rathna.				
Reference Books					
1.	A Textbook of Practical Chemistry by Dr. Sudarsan Barua				



BSC	MATHEMATICS-II (ODE & Complex Variables)	L	T	P	C
		3	1	0	4
Pre-requisite: Knowledge of Mathematics at Class XI & XII					
Course Objectives:					
1. To familiarize the prospective engineers with techniques in ordinary differential equations and complex variables 2. To provide the basic tools of mathematics for the purpose of modelling the problems and obtaining solutions.					
Course Outcome:					
After successful completion of the course, the students will learn CO 1: the effective mathematical tools for the solutions of differential equations that model physical processes. CO 2: the tools of differentiation and integration of functions of a complex variable that are used in various techniques dealing engineering problems.					
Module 1:First Order Ordinary Differential Equations					15hours
Exact, linear and Bernoulli's equations. Equations not of first degree: equations solvable for p, equations solvable for y, equations solvable for x and Clairaut's type.					
Module 2:Ordinary Differential Equations of Higher Orders					15 hours
Second order linear differential equations with variable coefficients: Euler-Cauchy equations, Solution by variation of parameters; Power series solutions: Legendre's equations and Legendre polynomials, Frobenius method, Bessel's equation and Bessel's functions of the first kind and their properties.					
Module 3: Complex Variable – Differentiation					15 hours
Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate; elementary analytic functions (exponential, trigonometric, logarithm) and their properties; Conformal mappings, Mobius transformations and their properties.					
Module 4: Complex Variable – Integration					15 hours
Contour integrals, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof), Liouville's theorem and Maximum-Modulus theorem (without proof); Taylor's series, zeros of analytic functions, singularities, Laurent's series; Residues, Cauchy Residue theorem (without proof), Evaluation of definite integral involving sine and cosine, Evaluation of certain improper integrals using the Bromwich contour.					
Total Lecture hours					60 hours
Text Book					
1. AICTE's Prescribed Textbook: Mathematics-II (Calculus, Ordinary Differential Equations and Complex Variable), Khanna Book Publishing Co.					
Reference Books					
1. ReenaGarg, Engineering Mathematics, Khanna Book Publishing Company, 2022. 2. ReenaGarg, Advanced Engineering Mathematics, Khanna Book Publishing Company, 2021. 3. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, John Wiley & Sons, 2006. 4. Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008. 5. W. E. Boyce and R. C. DiPrima, Elementary Differential Equations and Boundary Value Problems, 9th Edition, Wiley India, 2009 6. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005 7. S. L. Ross, Differential Equations, 3rd Edition, Wiley India, 1984 8. E. A. Coddington, An Introduction to Ordinary Differential Equations, Prentice Hall India, 1995. 9. E. L. Ince, Ordinary Differential Equations, Dover Publications, 1958					



ESC	PROGRAMMING FOR PROBLEM SOLVING	L	T	P	C
		3	0	0	3
Prerequisite: Basic computer knowledge, basic mathematics					
Course Objectives:					
1. To learn the fundamentals of computers. 2. To understand the various steps in program development. 3. To learn the syntax and semantics of C programming language. 4. To learn the usage of structured programming approach in solving problems. 5. To understated and formulate algorithm for programming script 6. To analyze the output based on the given input variable					
Course Outcome:					
After successful completion of the course, the students will learn CO1: Illustrate basic concepts of computer and C programming. CO2: Apply the concepts of conditional and looping statements. CO3: Demonstrate the ability to write C program using arrays, structures, pointers and files. CO4: Develop modular programs using C language.					
MODULE 1: Introduction to Programming					6 hours
Introduction to Programming; Introduction to components of a computer system (disks, memory, processor, where a program is stored and executed, operating system, compilers etc.) Idea of Algorithm: steps to solve logical and numerical problems. Representation of Algorithm: Flowchart/Pseudocode with examples. From algorithms to programs; source code, variables (with data types) variables and memory locations, Syntax and Logical Errors in compilation, object and executable code.					
MODULE 2: Introduction to C					6 hours
Using Comments, Keywords, Identifiers, Tokens, Basic Data Types, Writing C Expressions using Operators, Precedence of Operators, I/O Statements in C					
MODULE 3: Conditional Branching and Loops					6hours
Conditional Branching Statements, Iterative Statements, Nested Loops, Break and Continue Statements, Goto Statements.					
MODULE 4: Arrays and Strings					6 hours
1-D Array-Declaration, Accessing Array Elements, Array Operations, 2-D Array-Matrix Addition, Subtraction, Multiplication, Character Arrays, Strings, String Manipulation Function.					
MODULE 5: Functions					8 hours
Function Declaration/Prototype, Function Definition, Function Call, Return Statement, Passing Parameters, Scope of Variables, Storage Classes, Recursive Function. Example programs, such as Finding Factorial, Fibonacci series, Ackerman function etc. Quick sort or Merge sort.					
MODULE 6: Structure					7 hours
Structures, Defining Structures, Accessing Members, Array of Structures.					
MODULE 7: Pointers and File handling					6 hours
Pointers, Idea of pointers, Defining pointers, Use of Pointers in self-referential structures, File handling.					
Total hours					45 hours
Text Books					
1.	Byron Gottfried, Schaum's Outline of Programming with C, McGraw-Hill				
2.	Yashavant Kanetkar, Let us C, BPB Publication				
Reference Book					
Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India					



BSC	PROGRAMMING FOR PROBLEM SOLVING LABORATORY	L	T	P	C
		0	0	4	2
Prerequisite: Basic computer knowledge, Basic Mathematics					
Course Objectives:					

Internal Member

External Member/BoS



HSMC	ENGLISH FOR TECHNICAL WRITING	L	T	P	C
		2	0	2	3
Prerequisite: English language competence of 10+2 level					
Course Objectives: the objectives of this course are to:					
1.Provide learning environment to practice listening, speaking, reading and writing skills					
2.Assist the students to carry on the tasks and activities through guided instructions and materials					
3.Effectively integrate English language learning with employability skills and training					
4.Provide hands-on experience through case-studies, mini-projects, group and individual presentations.					
Course Outcome: After successful completion of this course, the students will be able to					
1. develop their basic as well as domain specific vocabulary					
2. apply the basic principles of effective writing in constructing meaningful sentences and paragraphs, and writing different styles of texts					
3. produce various academic and professional texts like essays, reports, and letters					
4. enhance their English language skills and employability skills through activities and training in a language laboratory					
Module 1: Vocabulary Building					8 hours
The concept of Word Formation, root words, prefixes and suffixes, synonyms, antonyms, and standard abbreviations,collocations, domain specific vocabulary used in real life contexts, vocabulary building exercises					
Module 2: Basic Writing Skills					8 hours
Mechanisms of writing: importance of proper punctuation, English punctuation marks, capitalization, semantic markers.Sentence Structures: simple, complex, compound. Use of phrases and clauses in sentences.Paragraphs: parts of a paragraph, topic sentence, supporting sentences, concluding sentence.Organizing principles of paragraphs, Creating coherence and unity, techniques for writing precisely					
Module 3: Nature and Style of sensible Writing					10 hours
Describing, defining classifying, providing examples or evidence, writing introduction and conclusion of a long text.					
Module 4: Identifying Common Errors in writing					4 hours
Subject-verb agreement, noun-pronoun agreement, misplaced modifiers, articles, prepositions, redundancies, clichés					
Module 5: Writing Practices					7 hours
Comprehension, formal letter writing, essay writing, report writing: features, types, format, structure, report writing process, sources of data collection, plagiarism.					
Module 6: Oral Communication (This Module involves interactive practice sessions in Language Lab)					8 hours
Listening Comprehension					
Pronunciation, Intonation, Stress and Rhythm					
Common Everyday Situations: Conversations and Dialogues					
Communication at Workplace					
Interviews					
Formal Presentations					
Total hours					45 hours
Text Book					
1.	AICTE’s Prescribed Textbook: English (with Lab Manual) ISBN: 978-93-91505-097				
Reference Books					



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| 1. | Effective Communication Skills. Kul Bhushan Kumar, Khanna Book Publishing, 2022. |
| 2. | Practical English Usage. Michael Swan. OUP. 1995. |
| 3. | Remedial English Grammar. F.T. Wood. Macmillan. 2007 |
| 4. | On Writing Well. William Zinsser. Harper Resource Book. 2001 |
| 5. | Study Writing. Liz Hamp- Lyons and Ben Heasley. Cambridge University Press. 2006. |
| 6. | Communication Skills. Sanjay Kumar and PushpLata. Oxford University Press. 2011. |
| 7. | Exercises in Spoken English. Parts. I-III. CIEFL, Hyderabad. Oxford University Press. |

Internal Member

External Member/BoS



HSMC	Universal Human Values-II: Understanding Harmony And Ethical Human Conduct	L	T	P	C
		2	1	0	3
Prerequisite: UHV 1 / SIP					
Course Objectives:					
This introductory course input is intended: 1. To help the students appreciate the essential complementarity between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings. 2. To facilitate the development of a Holistic perspective among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of the Human reality and the rest of existence. Such a holistic perspective forms the basis of Universal Human Values and movement towards value-based living in a natural way. 3. To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behavior and mutually enriching interaction with Nature.					
Course Outcome:					
By the end of the course, students are expected to become more aware of themselves, and their surroundings (family, society, nature); they would become more responsible in life, and in handling problems with sustainable solutions, while keeping human relationships and human nature in mind. They would have better critical ability. They would also become sensitive to their commitment towards what they have understood (human values, human relationship and human society). It is hoped that they would be able to apply what they have learnt to their own self in different day-to-day settings in real life, at least a beginning would be made in this direction.					
Module:1 Introduction					9 hours
Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) , Understanding Value Education ,Sharing about Oneself ,Self-exploration as the Process for Value Education , Continuous Happiness and Prosperity – the Basic Human Aspirations , Exploring Human Consciousness ,Happiness and Prosperity – Current Scenario , Method to Fulfill the Basic Human Aspirations ,Exploring Natural Acceptance					
Module:2 Harmony in the Human Being					9 hours
Understanding Human being as the Co-existence of the Self and the Body, Distinguishing between the Needs of the Self and the Body, Exploring the difference of Needs of Self and Body. The Body as an Instrument of the Self. Understanding Harmony in the Self. Exploring Sources of Imagination in the Self. Harmony of the Self with the Body. Programme to ensure self-regulation and Health. Exploring Harmony of Self with the Body					
Module:3 Harmony in the Family and Society					9 hours
Harmony in the Family – the Basic Unit of Human Interaction. Trust' – the Foundational Value in Relationship. Exploring the Feeling of Trust. 'Respect' – as the Right Evaluation.Exploring the Feeling of Respect.Other Feelings, Justice in Human-to-Human Relationship. Understanding Harmony in the Society. Vision for the Universal Human Order. Exploring Systems to fulfil Human Goal.					
Module:4 Harmony in the Nature/Existence					9 hours
Understanding Harmony in the Nature. Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature. Exploring the Four Orders of Nature. Realizing Existence as Co-existence at All Levels. The Holistic Perception of Harmony in Existence Exploring Co-existence in Existence.					



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Module:5 Implications of the Holistic Understanding		9 hours
Natural Acceptance of Human Values. Definitiveness of (Ethical) Human Conduct. Exploring Ethical Human Conduct. A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order. Competence in Professional Ethics. Exploring Humanistic Models in Education. Holistic Technologies, Production Systems and Management Models-Typical Case Studies. Strategies for Transition towards Value-based Life and Profession. Exploring Steps of Transition towards Universal Human Order.		
Total hours		45 hours
Text Book		
1.	A Foundation Course in Human Values and Professional Ethics, R R Gaur, R Asthana, G P Bagaria, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-47	
Reference Books		
1.	Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.	
2.	Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.	
3.	The Story of Stuff (Book).	
4.	The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi.	

Internal Member

External Member/BoS



ESC	MANUFACTURING PRACTICE WORKSHOP	L	T	P	C
		0	0	4	2
Prerequisite: None					
Course Objectives: The objectives of this course are to:					
1. To impart knowledge and skill to use tools, machines, equipment, and measuring instruments. 2. To educate students of safe handling of machines and to develop the hands-on practical workshop skills.					
Course Outcome: After successful completion of this course, the students will be able to					
1. Select tools and machinery according to the job. 2. Use hand tools in different shops for performing different operations. 3. Prepare job according to the drawing.					
Module 1: Welding					5 hours
(a) Theoretical Instructions: Introduction to welding processes, Safety Precautions, Demonstration of different equipments, tools used in welding, various fluxes & electrodes used in welding. Introduction of AC & DC welding and its applications.					
(b) Practical Demonstrations: Demonstration of all basic tools & personal protective equipments. Demonstration of operations such as measuring, marking, punching and cutting. Demonstration of different types of joints by using arc welding, gas welding and flame brazing.					
Module 2: Turning					5 hours
(a) Theoretical Instructions: Introduction of machine and machine tools, Safety Precautions, Different equipments and tools used, basic metal cutting operations. Introduction of various types of cutting tools (Nomenclature) and their material.					
(b) Practical Demonstrations: Demonstration on Lathe & basic operations such as drilling, facing, turning, taper turning, step turning, knurling, chamfering, threading. Demonstration of basic measuring instruments.					
Module 3: Machining					5 hours
(a) Theoretical Instructions: Introduction to machine tools such as milling, shaper and surface grinder. Safety Precautions. Demonstration of different tools and material used, Demonstration of basic measuring instruments used.					
(b) Practical Demonstrations: Demonstration on basic operations such as gear cutting, hexagonal bolt, grinding, slot cutting and fitting.					
Module 4: Fitting					5 hours
(a) Theoretical Instructions: Introduction to fitting work, safety precautions, Demonstration of basic hand tools, holding devices and basic fitting operations such as measuring, marking, punching, filing, sawing, drilling, tapping and dieing.					
(b) Practical Demonstrations: Demonstration of all basic hand tools, measuring tools & equipments. Demonstration of simple operations such as marking, measuring, punching, filing, sawing, drilling, tapping and dieing.					
Module 5: Carpentry					5 hours
(a) Theoretical Instructions: Introduction to Carpentry, Safety Precautions, demonstration of different tools used in carpentry. Various types of joints. Brief description of wood cutting machines.					
(b) Practical Demonstrations: Demonstration & practice of different carpentry operation like marking and measuring, cutting, planning, chiseling, filing and chamfering.					
Module 6: Blacksmithy					5 hours
(a) Theoretical Instructions: Introduction, Safety precautions, Demonstration of basic hand					



tools and holding devices, Description of all forging operations such as heating, hammering, finishing, forge welding, normalizing and tempering. Comparison of hot & cold working.

(b) Practical Demonstrations: Demonstration & practice of different smithy operations like cutting, hammering, punching, bending etc. Demonstration & practice of making a square dimension from a cylindrical bar and vice versa. Total Hours : 30 hours

Text Book(s)

1. Hajra Choudhury S.K., Hajra Choudhury A.K. and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010.
2. Kalpakjian S, Steven S. Schmid, "Manufacturing Engineering and Technology", Pearson Education India Edition, 4th Edition, 2002
3. Rao P.N., "Manufacturing Technology", Vol. I and Vol. II, Tata McGraw-Hill House, 2017

Reference Book

1. Workshop Practice – Singh S., S.K. Kataria & Sons. 2003.

Internal Member

External Member/BoS



AU	Sports and Yoga	L	T	P	C
		2	0	0	0
Prerequisite: Nil					
Course Objectives:					
1. To make the students understand the importance of sound health and fitness principles as they relate to better health. 2. To expose the students to a variety of physical and yogic activities aimed at stimulating their continued inquiry about Yoga, physical education, health and fitness 3. To create a safe, progressive, methodical and efficient activity based plan to enhance improvement and minimize risk of injury. 4. To develop among students an appreciation of physical activity as a lifetime pursuit and a means to better health.					
Course Outcome:					
On successful completion of the course the students will be able: 1. To practice Physical activities and Hatha Yoga focusing on yoga for strength, flexibility, and relaxation. 2. To learn techniques for increasing concentration and decreasing anxiety which leads to stronger academic performance. 3. To learn breathing exercises and healthy fitness activities 4. To understand basic skills associated with yoga and physical activities including strength and flexibility, balance and coordination. 5. To perform yoga movements in various combination and forms. 6. To assess current personal fitness levels. 7. To identify opportModuleies for participation in yoga and sports activities. 8. To develop understanding of health-related fitness components: cardiorespiratory endurance, flexibility and body composition etc. 9. To improve personal fitness through participation in sports and yogic activities. 10. To develop understanding of psychological problems associated with the age and lifestyle. 11. To demonstrate an understanding of sound nutritional practices as related to health and physical performance. 12. To assess yoga activities in terms of fitness value. 13. To identify and apply injury prevention principles related to yoga and physical fitness activities. 14. To understand and correctly apply biomechanical and physiological principles elated to exercise and training.					
Module:1 Introduction to Physical Education					
Meaning & definition of Physical Education Aims & Objectives of Physical Education Changing trends in Physical Education					
Module:2 Olympic Movement					
Ancient & Modern Olympics (Summer & Winter) Olympic Symbols, Ideals, Objectives & Values Awards and Honours in the field of Sports in India (Dronacharya Award, Arjuna Award, Dhayanchand Award, Rajiv Gandhi Khel Ratna Award etc.)					
Module:3 Physical Fitness, Wellness & Lifestyle					
Meaning & Importance of Physical Fitness & Wellness Components of Physical fitness Components of Health related fitness					



Components of wellness o Preventing Health Threats through Lifestyle Change
Concept of Positive Lifestyle

Module:4 Fundamentals of Anatomy & Physiology in Physical Education, Sports and Yoga

Define Anatomy, Physiology & Its Importance
Effect of exercise on the functioning of Various Body Systems. (Circulatory System, Respiratory System, Neuro-Muscular System etc.)

Module:5 Kinesiology, Biomechanics & Sports

Meaning & Importance of Kinesiology & Biomechanics in Physical Edu. & Sports
Newton's Law of Motion & its application in sports.
Friction and its effects in Sports.

Module: 6 Postures

Meaning and Concept of Postures.
Causes of Bad Posture.
Advantages & disadvantages of weight training.
Concept & advantages of Correct Posture.
Common Postural Deformities – Knock Knee; Flat Foot; Round Shoulders; Lordosis, Kyphosis, Bow Legs and Scoliosis.
Corrective Measures for Postural Deformities

Module: 7 Yoga

Meaning & Importance of Yoga
Elements of Yoga o Introduction - Asanas, Pranayama, Meditation & Yogic Kriyas
Yoga for concentration & related Asanas (Sukhasana; Tadasana; Padmasana & Shashankasana)
Relaxation Techniques for improving concentration - Yog-nidra

Module: 8 Yoga & Lifestyle

Asanas as preventive measures.
Hypertension: Tadasana, Vajrasana, Pavan Muktasana, Ardha Chakrasana, Bhujangasana, Sharasana.
Obesity: Procedure, Benefits & contraindications for Vajrasana, Hastasana, Trikonasana, Ardh Matsyendrasana.
Back Pain: Tadasana, Ardh Matsyendrasana, Vakrasana, Shalabhasana, Bhujangasana.
Diabetes: Procedure, Benefits & contraindications for Bhujangasana, Paschimottasana, Pavan Muktasana, Ardh Matsyendrasana.
Asthema: Procedure, Benefits & contraindications for Sukhasana, Chakrasana, Gomukhasana, Parvatasana, Bhujangasana, Paschimottasana, Matsyasana.

Module: 9 Training and Planning in Sports

Meaning of Training
Warming up and limbering down
Skill, Technique & Style
Meaning and Objectives of Planning.
Tournament – Knock-Out, League/Round Robin & Combination.

Module:10 Psychology & Sports

Definition & Importance of Psychology in Physical Edu. & Sports
Define & Differentiate Between Growth & Development
Adolescent Problems & Their Management
Emotion: Concept, Type & Controlling of emotions
Meaning, Concept & Types of Aggressions in Sports.
Psychological benefits of exercise.



Anxiety & Fear and its effects on Sports Performance.

Motivation, its type & techniques.

Understanding Stress & Coping Strategies.

Module:11 Doping

Meaning and Concept of Doping

Prohibited Substances & Methods

Side Effects of Prohibited Substances

Module:12 Sports Medicine

First Aid – Definition, Aims & Objectives.

Sports injuries: Classification, Causes & Prevention.

Management of Injuries: Soft Tissue Injuries and Bone & Joint Injuries

Module:13 Sports / Games

Following subtopics related to any one Game/Sport of choice of student out of: Athletics, Badminton, Basketball, Chess, Cricket, Kabaddi, Lawn Tennis, Swimming, Table Tennis, Volleyball, Yoga etc.

History of the Game/Sport.

Latest General Rules of the Game/Sport.

Specifications of Play Fields and Related Sports Equipment.

Important Tournaments and Venues.

Sports Personalities.

Proper Sports Gear and its Importance.

Total Hours: 30 Hours

Text Book

1. Modern Trends and Physical Education by Prof. Ajmer Singh

Reference Books

1. Light On Yoga by B.K.S. Iyengar.
2. Health and Physical Education – NCERT (11th and 12th Classes)

Internal Member

External Member/BoS